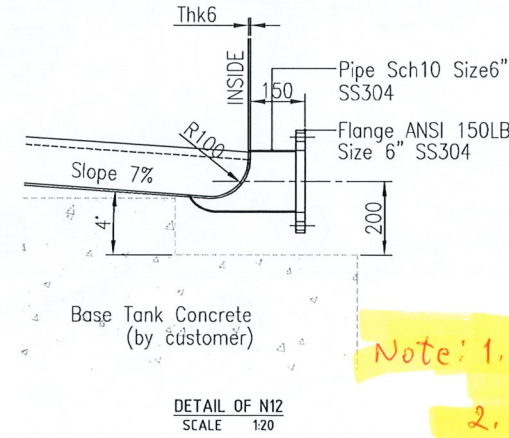
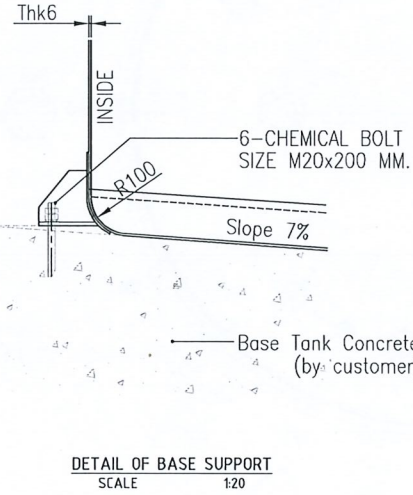
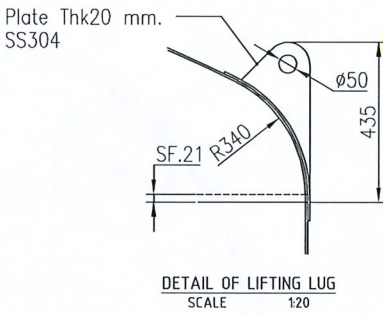
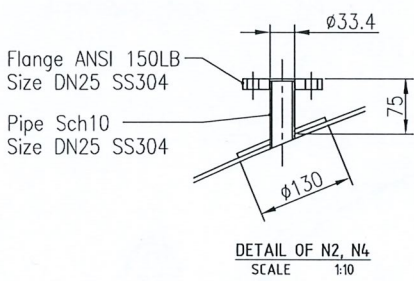
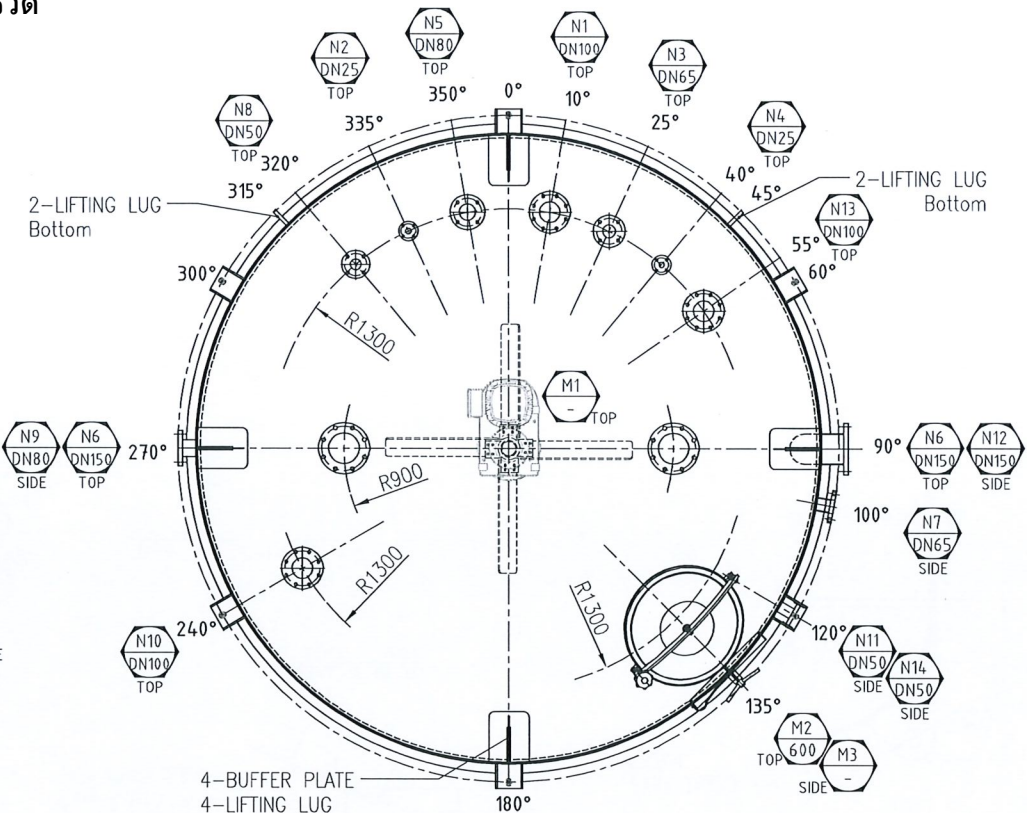
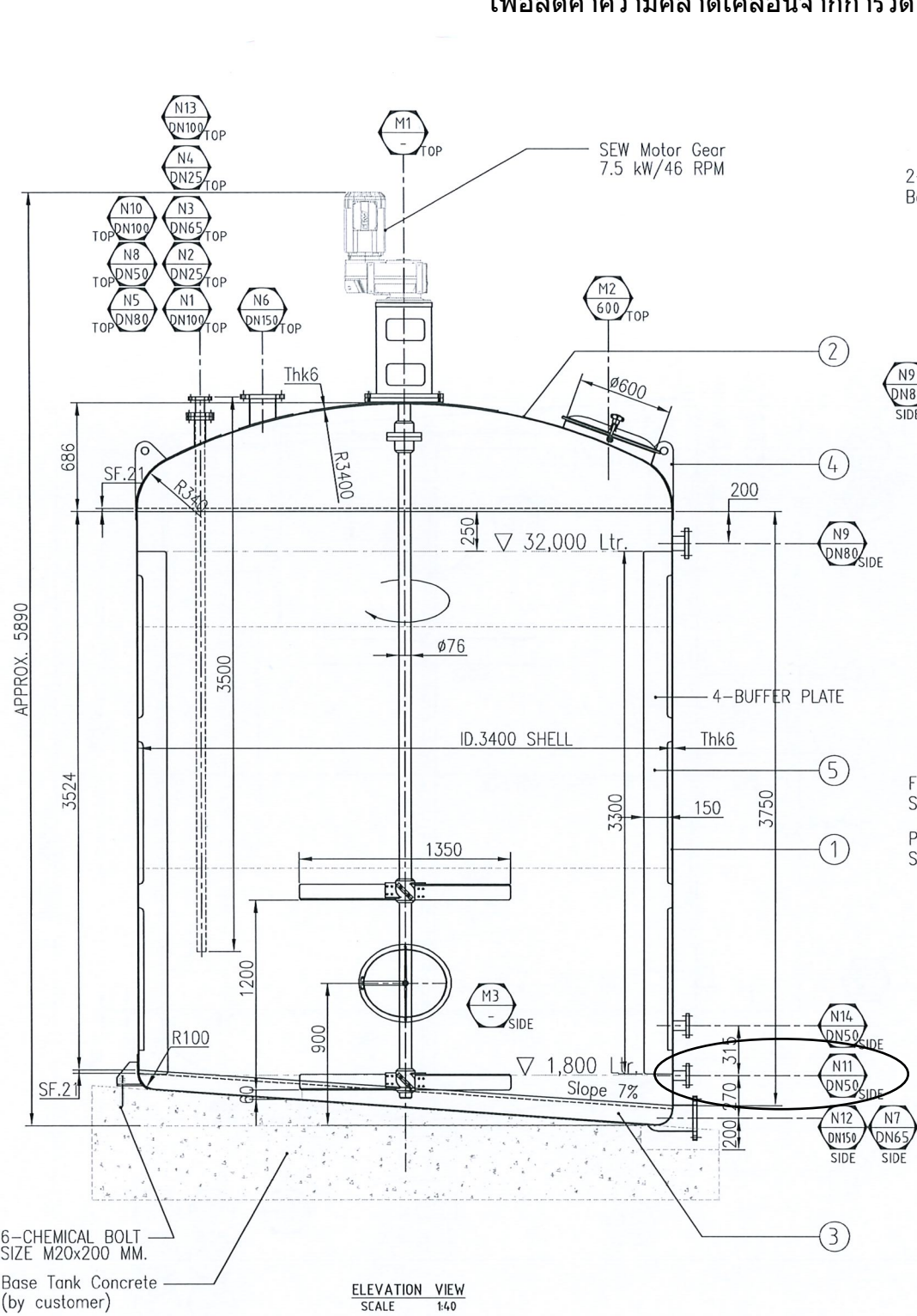


N12 และ N14 จะเป็น Weld-in Socket

ตำแหน่งของ N12(Level) กับ N7(outlet)
ห่างกันอย่างน้อย 2 ฟุต หรือ 600mm
N14(pH)ติดให้ใกล้จุดทาง outletหรือจุดที่เก็บตัวอย่าง
เพื่อลดค่าความคลาดเคลื่อนจากการวัด

ALL NOZZLE POSITION ACCORDING
TO TOP VIEW AND BOTTOM VIEW



DESIGN DATA		
DESIGN CODE	ASME, SECTION VIII DIV.1, 2023 ED (APPLY)	
CODE STAMP (U-STAMP)	NO	
TYPE OF HEAD	TORISPHERICAL 1:10	
TYPE OF SHELL	CYLINDRICAL	
LIQUID NAME	SHELL	JACKET
	Adjustment Tank	-
DENSITY	kg/m ³	1170
CORROSION ALLOWANCE	MM.	0
RADIOGRAPHED		None
JOINT EFFICIENCY (SHELL/HEAD)		0.7/0.7
OPERATING TEMPERATURE	°C	40
DESIGN TEMPERATURE	°C	70
MDMT.	°C	-
OPERATING PRESSURE	bar(g)	ATM
DESIGN PRESSURE	bar(g)	1
HYDROSTATIC TEST PRESSURE	bar(g)	No
PNEUMATIC TEST PRESSURE	bar(g)	No
INSULATION		-
PAINTING		None

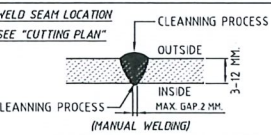
MATERIAL SPECIFICATION		
SHELL	ASTM A240 TP.304	
TOP HEAD	ASTM A240 TP.304	
JACKET (DIMPLE PLATE)	-	
MANHOLE AND NOZZLE	STAINLESS 304	
GASKET	-	
EXTERNAL (BOLT+NUT) / INTERNAL (BOLT+NUT)	STAINLESS STEEL 304 / STAINLESS STEEL 304	

CAPACITY AND WEIGHT		
FLUID WEIGHT	kg.	-
OPERATING CAPACITY	Liter.	32,000
FULL CAPACITY	Liter.	34,000
EMPTY WEIGHT	kg.	5,200
OPERATING WEIGHT	kg.	37,200

NOZZLE SCHEDULE							
NOZZLE	DESCRIPTION	SIZE	Q'TY	NOZZLE NECK SCH/THK	MATERIAL	FLANGE / COUPLING CLASS	TYPE
M1	Agitator	-	1	-	SS316	-	-
M2	Top Manhole	ø600	1	6I	SS304	MANHOLE	SS304
M3	Side Manhole	-	1	6I	SS304	MANHOLE	SS304
N1	Slurry Inlet	DN100	1	10S.	SS304	FLANGE	SS304 AISI150
N2	5% NaOH Inlet	DN25	1	10S.	SS304	FLANGE	SS304 AISI150
N3	35% HCl Inlet (Deep Pipe)	DN65	1	10S.	SS304	FLANGE	SS304 AISI150
N4	3% NO2SO Inlet	DN25	1	10S.	SS304	FLANGE	SS304 AISI150
N5	Water Inlet	DN80	1	10S.	SS304	FLANGE	SS304 AISI150
N6	CIP Spray Ball	DN150	2	10S.	SS304	FLANGE	SS304 AISI150
N7	Slurry Outlet	DN65	1	10S.	SS304	FLANGE	SS304 AISI150
N8	Slurry Circulation	DN50	1	10S.	SS304	FLANGE	SS304 AISI150
N9	Over Flow	DN80	1	10S.	SS304	FLANGE	SS304 AISI150
N10	Vent	DN100	1	10S.	SS304	FLANGE	SS304 AISI150
N11	Level Transmitter	DN50	1	10S.	SS304	FLANGE	SS304 AISI150
N12	Drain	DN150	1	10S.	SS304	FLANGE	SS304 AISI150
N13	Spare	DN100	1	10S.	SS304	FLANGE	SS304 AISI150
N14	pH Probe	DN25, DN50	1	10S.	SS304	FLANGE	SS304 AISI150

BILL OF MATERIAL			
NO.	Q'TY	DESCRIPTION	MATERIAL
1	1	SHELL	PL 6 MM.THK. ASTM A240 TP304
2	1	TOP HEAD	PL 6 MM.THK. ASTM A240 TP304
3	1	BOTTOM HEAD	PL 6 MM.THK. ASTM A240 TP304
4	4	LIFTING LUG	PL 20 MM.THK. ASTM A240 TP304
5	4	BUFFER PLATE	PL 6 MM.THK. ASTM A240 TP304

FOR APPROVAL
Date.....



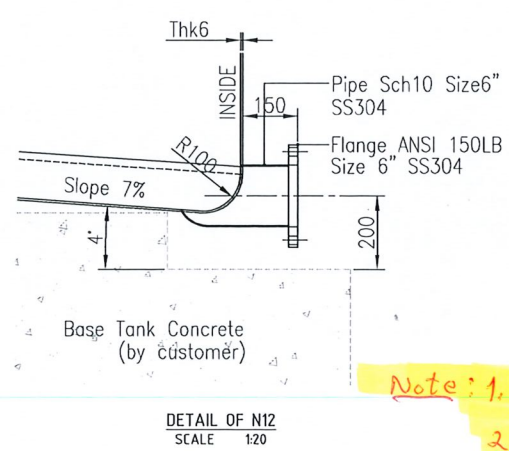
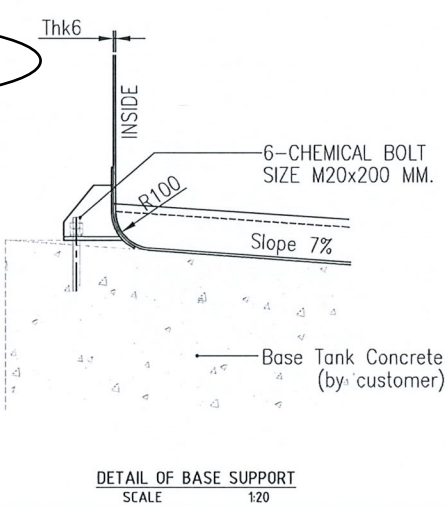
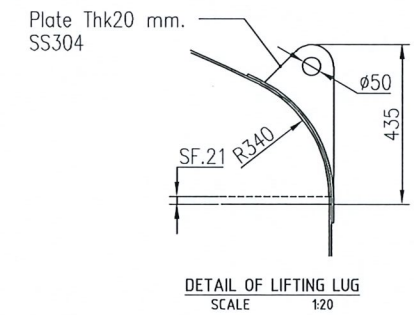
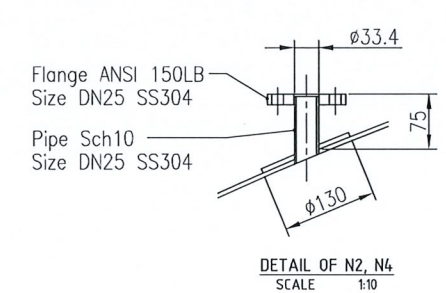
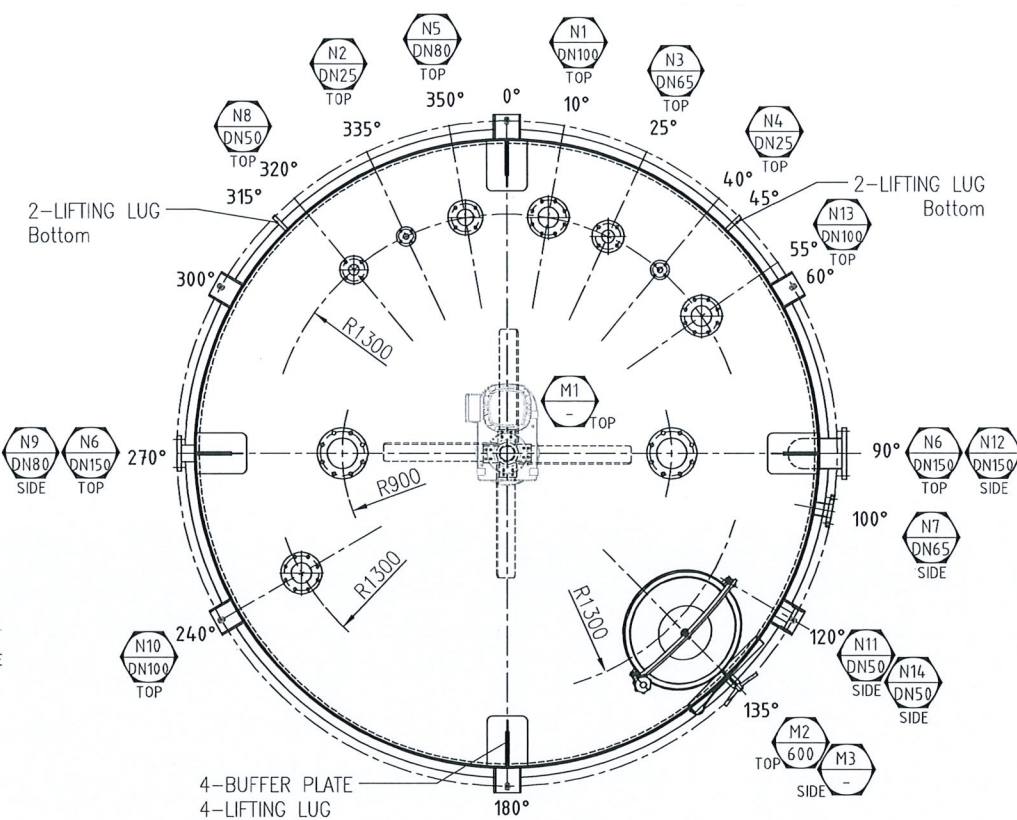
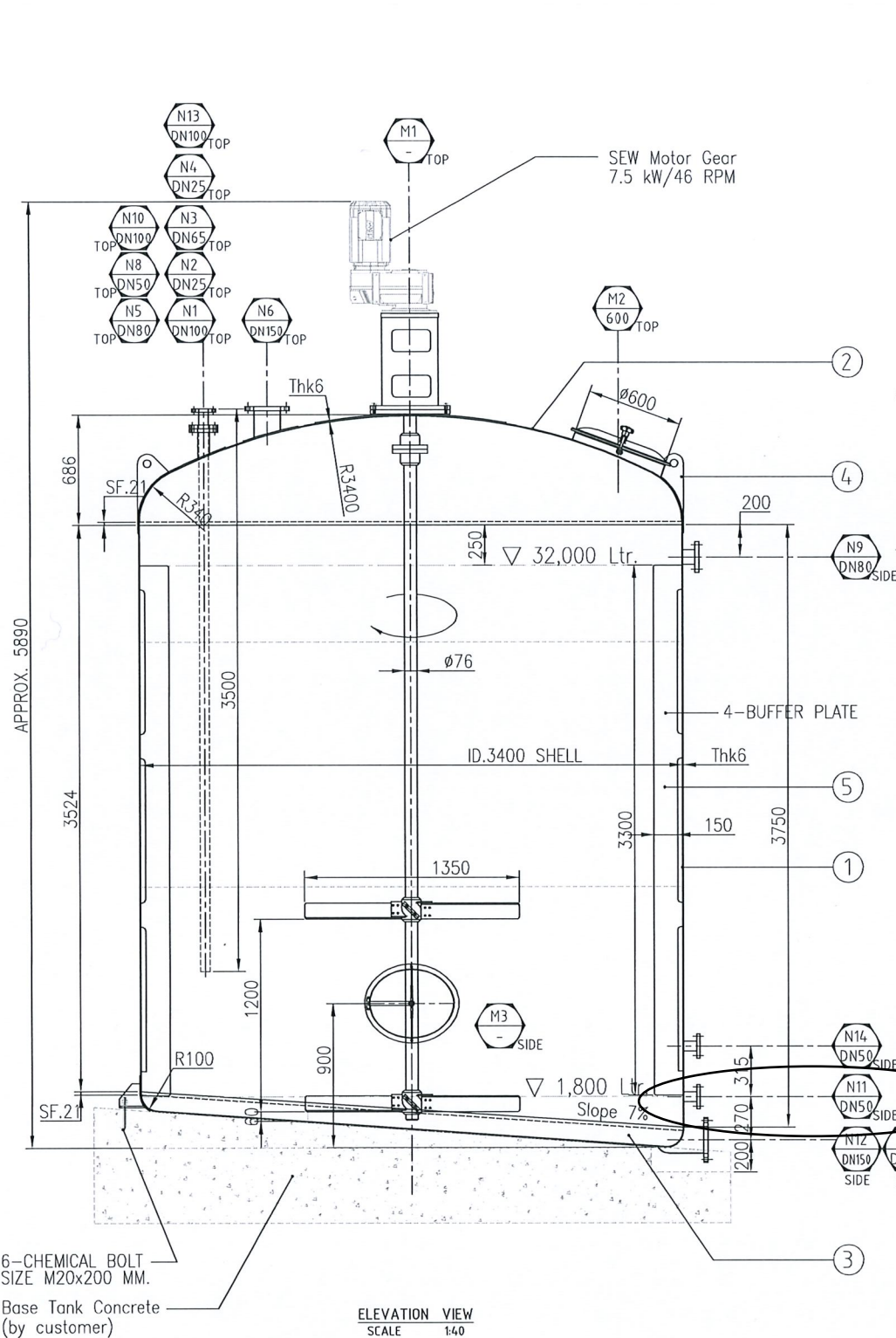
COMPACT INNOVATION CO.,LTD.
168/40 Moo 1, Lumpukud, Tanyaburi, Phathumtani, 12110, Thailand
Tel. +66-2957 2326-7 Fax +66-2957 2328 Email: compact@cpi.co.th

DRAWING BY:
Thammanoon P.
APPROVED BY:
Numphon N.
DRAWING DATE:
4/02/2026

OWNER:
Ingredion (Thailand) Co., Ltd.
PROJECT:
Sprint 18K Spray Dryer
TITLE:
TS-40101 pH Adjustment Tank 32 m3

DRAWING NO.
CIC-IGD-26001-ME-DWG-001
SCALE:
1:40
SHEET NO.
1/1
FIRST ANGLE PROJECTION
SIZE: A3

REVISION	DESCRIPTION	DATE
1		
2		
3		
4		
5		



DESIGN DATA		
DESIGN CODE	ASME SECTION VIII DIV.1, 2023 ED (APPLY)	
CODE STAMP (U-STAMP)	NO	
TYPE OF HEAD	TORISPHERICAL 1:10	
TYPE OF SHELL	CYLINDRICAL	
LIQUID NAME	SHELL	JACKET
	Adjustment Tank	-
DENSITY	kg/m ³	1170
CORROSION ALLOWANCE	MM	0
RADIOGRAPHED		None
JOINT EFFICIENCY (SHELL/HEAD)		0.7/0.7
OPERATING TEMPERATURE	°C	40
DESIGN TEMPERATURE	°C	70
MDMT	°C	-
OPERATING PRESSURE	bar(g)	ATM
DESIGN PRESSURE	bar(g)	1
HYDROSTATIC TEST PRESSURE	bar(g)	No
PNEUMATIC TEST PRESSURE	bar(g)	No
INSULATION		
PAINTING	None	

MATERIAL SPECIFICATION	
SHELL	ASTM A240 TP304
TOP HEAD	ASTM A240 TP304
JACKET (DIMPLE PLATE)	-
MANHOLE AND NOZZLE	STAINLESS 304
GASKET	-
EXTERNAL (BOLT+NUT) / INTERNAL (BOLT+NUT)	STAINLESS STEEL 304 / STAINLESS STEEL 304

CAPACITY AND WEIGHT	
FLUID WEIGHT	kg
OPERATING CAPACITY	Liter
FULL CAPACITY	Liter
EMPTY WEIGHT	kg
OPERATING WEIGHT	kg

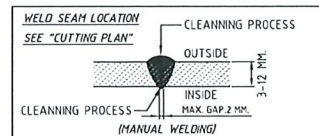
NOZZLE SCHEDULE							
NOZZLE	DESCRIPTION	SIZE	Q'TY	NOZZLE NECK SCH/THK	MATERIAL	FLANGE / COUPLING CLASS	TYPE
M1	Agitator	-	1	-	SS316	-	-
M2	Top Manhole	600	1	6T	SS304	MANHOLE	SS304
M3	Side Manhole	-	1	6T	SS304	MANHOLE	SS304
N1	Slurry Inlet	DN100	1	10S	SS304	FLANGE	SS304 AISI150
N2	5% NaOH Inlet	DN25	1	10S	SS304	FLANGE	SS304 AISI150
N3	35% HCl Inlet (Deep Pipe)	DN65	1	10S	SS304	FLANGE	SS304 AISI150
N4	3% NO2SO Inlet	DN25	1	10S	SS304	FLANGE	SS304 AISI150
N5	Water Inlet	DN80	1	10S	SS304	FLANGE	SS304 AISI150
N6	CIP Spray Ball	DN150	2	10S	SS304	FLANGE	SS304 AISI150
N7	Slurry Outlet	DN65	1	10S	SS304	FLANGE	SS304 AISI150
N8	Slurry Circulation	DN50	1	10S	SS304	FLANGE	SS304 AISI150
N9	Over Flow	DN80	1	10S	SS304	FLANGE	SS304 AISI150
N10	Vent	DN100	1	10S	SS304	FLANGE	SS304 AISI150
N11	Level Transmitter	DN50	1	10S	SS304	FLANGE	SS304 AISI150
N12	Drain	DN150	1	10S	SS304	FLANGE	SS304 AISI150
N13	Spare	DN100	1	10S	SS304	FLANGE	SS304 AISI150
N14	pH Probe	DN50	1	10S	SS304	FLANGE	SS304 AISI150

BILL OF MATERIAL			
NO.	Q'TY	DESCRIPTION	MATERIAL
1	1	SHELL	PL.6 MM.THK. ASTM A240 TP304
2	1	TOP HEAD	PL.6 MM.THK. ASTM A240 TP304
3	1	BOTTOM HEAD	PL.6 MM.THK. ASTM A240 TP304
4	4	LIFTING LUG	PL.20 MM.THK. ASTM A240 TP304
5	4	BUFFER PLATE	PL.6 MM.THK. ASTM A240 TP304

NOTE:

- VESSEL SURFACE
 - @ INSIDE SURFACE FLAT GRINDING AND POLISHING PROCESS Ra <=0.4 MICRON
 - @ OUTSIDE SURFACE (2B) FLAT GRINDING AND POLISHING PROCESS Ra <=0.8 MICRON
- ALL DIMENSIONS ARE MM. EXCEPT OTHERWISE SPECIFIED.
- FABRICATION TOLERANCE FOR SHELL AND HEAD ACCORDS TO ASME VII DIV.1 PER UG80, UG81.
- WELDS SHALL BE NEAT IN APPEARANCE, FREE FROM SLAGS, UNDERCUTS AND OTHER DEFECTS.
- VESSEL SHALL BE CLEANED OF SCALE, OIL, WELD SPATTER AND ALL OTHER FOREIGN MATTER BEFORE HYDROSTATIC TESTING.
- FOR NAME LETTERS, FRAME AND STAMPING SQUARES IN THE NAMEPLATE SHALL BE EMBOSSED AND THEIR FACES SHALL BE POLISHED BY BUFFING.
- FOR WELDING SYMBOL : F.P = FULL THICKNESS PENETRATION H.P = HALF THICKNESS PENETRATION

FOR APPROVAL
Date.....



COMPACT INNOVATION CO.,LTD.
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DRAWING BY :
Thammanoon P.
APPROVED BY :
Numphon N.
DRAWING DATE :
4/02/2026

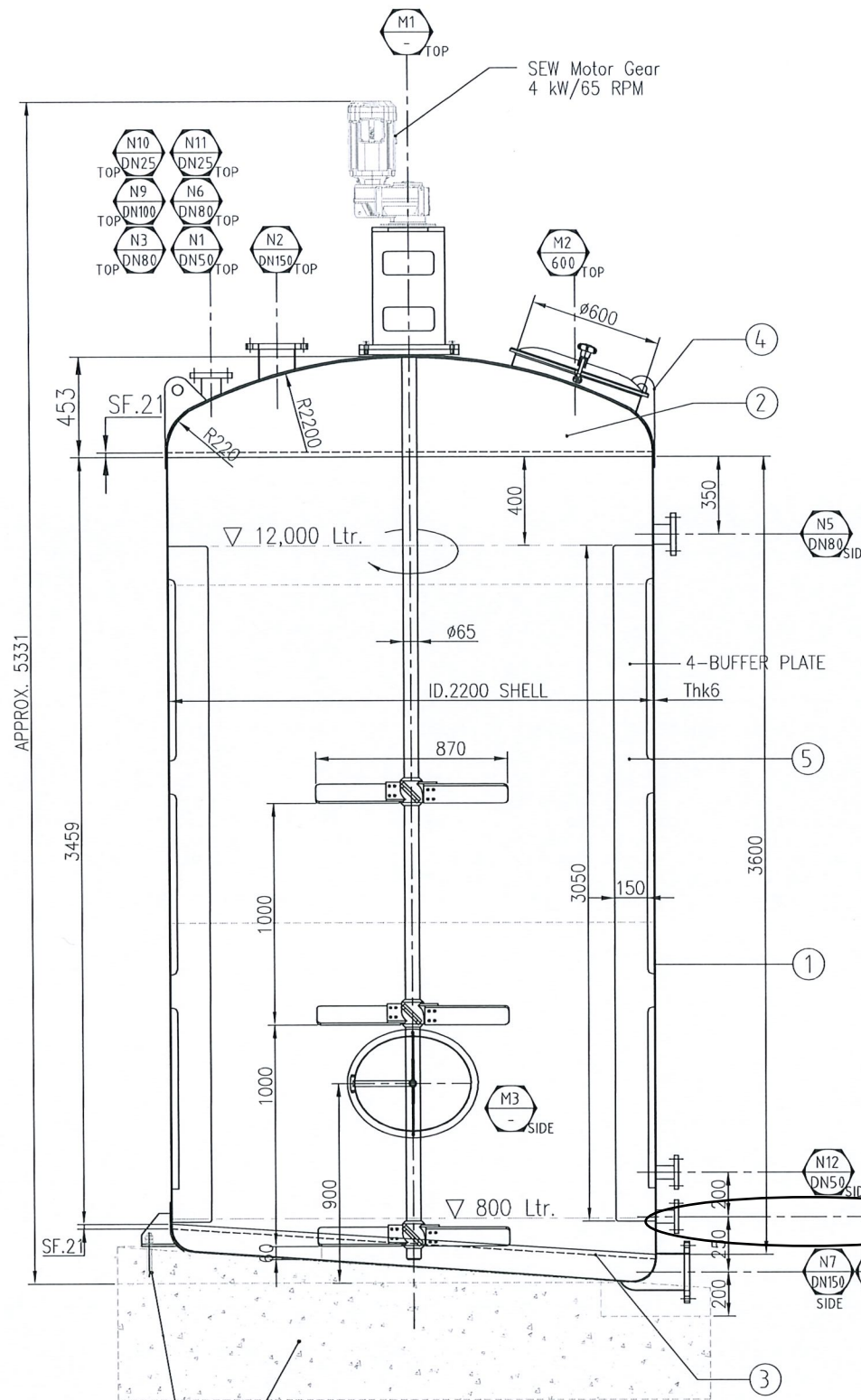
OWNER :
Ingredion (Thailand) Co., Ltd.
PROJECT :
Sprint 18K Spray Dryer
TITLE :
TS-40102 pH Adjustment Tank 32 m3

DRAWING NO.
CIC-IGD-26001-ME-DWG-002
SCALE :
1:40
SHEET NO.
1/1
FIRST ANGLE PROJECTION
SIZE : A3

REVISION	DESCRIPTION	DATE

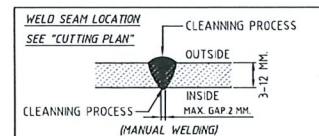
Note: 1. DRD Flange (supply by owner) Nozzle Length=0
2. Weld-in Socket (supply by owner) Nozzle Length=0

APPROX. 5331



6-CHEMICAL BOLT
SIZE M20x200 MM.
Base Tank Concrete
(by customer)

FOR APPROVAL
Date.....

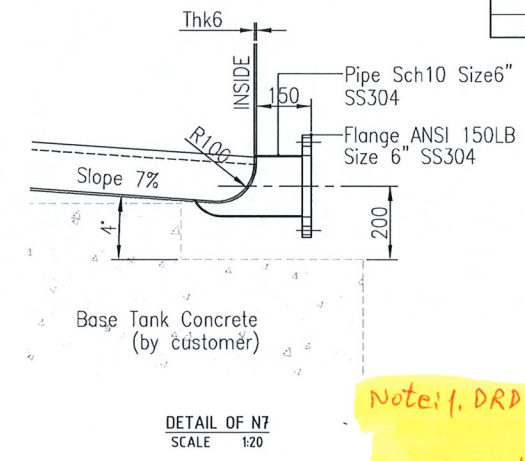
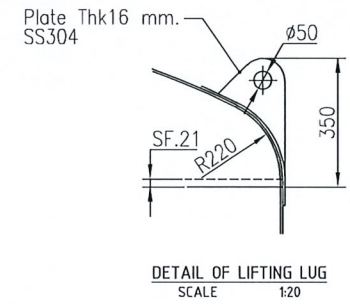
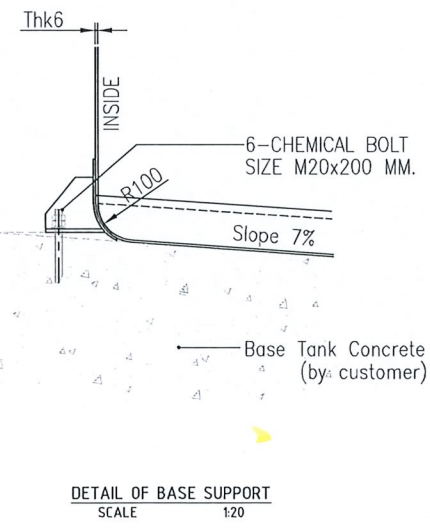
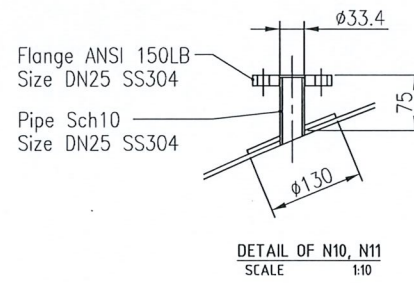
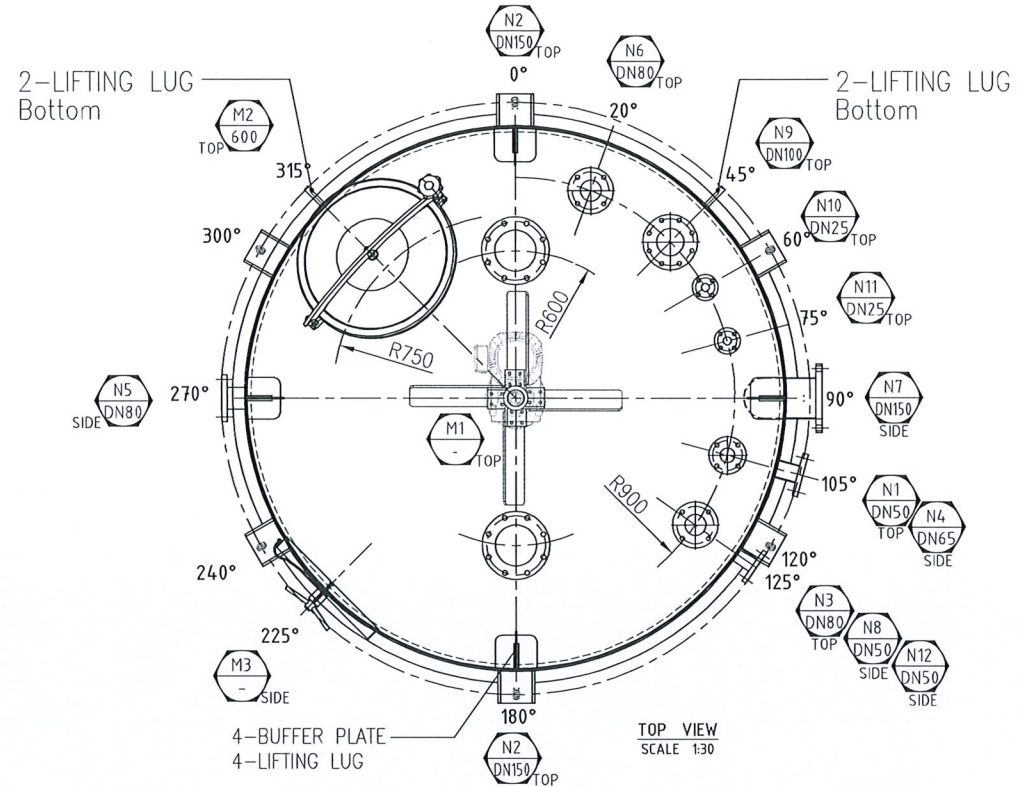


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168/40 Moo 1, Lumpukkud, Tanyaburi, Phathumtani, 12110, Thailand
Tel. +66-2957 2326-7 Fax +66-2957 2328 Email: compact@cpil.co.th

DRAWING BY: **Thammanoon P.**
APPROVED BY: **Numphon N.**
DRAWING DATE: **4/02/2026**
OWNER: **Ingredion (Thailand) Co., Ltd.**
PROJECT: **Sprint 18K Spray Dryer**
TITLE: **TS-40201 Spray Dryer Infeed Tank 12 m3**

DRAWING NO.		DESCRIPTION		DATE	REVISION
CIC-IGD-26001-ME-DWG-003					
SCALE :	SHEET NO.				
1:30	1/1				
FIRST ANGLE PROJECTION	SIZE : A3				
					

ALL NOZZLE POSITION ACCORDING
TO TOP VIEW AND BOTTOM VIEW



Note 1. DRD Flange (supply by owner) Nozzle = 0
2. Weld-in socket (SS304, long 40mm) Nozzle length = 0

DETAIL OF N3, N6
SCALE 1:15

Flange ANSI 150LB
Size DN80 SS304
Pipe Sch10
Size DN80 SS304

DESIGN DATA

DESIGN CODE		ASME SECTION VIII DIV.1, 2023 ED (APPLY)	
CODE STAMP (U-STAMP)		NO	
TYPE OF HEAD		TORISPHERICAL 1:10	
TYPE OF SHELL		CYLINDRICAL	
		SHELL	JACKET
LIQUID NAME		Spray Dryer Infeed Tank	
DENSITY	kg/m ³	1170	-
CORROSION ALLOWANCE	MM	0	-
RADIOGRAPHED		None	-
JOINT EFFICIENCY (SHELL/HEAD)		0.7/0.7	-
OPERATING TEMPERATURE	°C	40	-
DESIGN TEMPERATURE	°C	70	-
MDMT.	°C	-	-
OPERATING PRESSURE	bar(g)	ATM	-
DESIGN PRESSURE	bar(g)	1	-
HYDROSTATIC TEST PRESSURE	bar(g)	No	-
PNEUMATIC TEST PRESSURE	bar(g)	No	-
INSULATION		-	
PAINTING		None	

MATERIAL SPECIFICATION

SHELL	ASTM A240 TP.304
TOP HEAD	ASTM A240 TP.304
JACKET (DIMPLE PLATE)	-
MANHOLE AND NOZZLE	STAINLESS 304
GASKET	-
EXTERNAL (BOLT+NUT) / INTERNAL (BOLT+NUT)	STAINLESS STEEL 304 / STAINLESS STEEL 304

CAPACITY AND WEIGHT

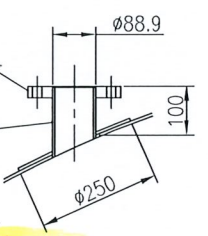
FLUID WEIGHT	kg.	-
OPERATING CAPACITY	Liter.	12,000
FULL CAPACITY	Liter.	13,700
EMPTY WEIGHT	kg.	3,300
OPERATING WEIGHT	kg.	15,300

NOZZLE SCHEDULE

NOZZLE	DESCRIPTION	SIZE	Q'TY	NOZZLE NECK SCH/THK	MATERIAL	FLANGE / COUPLING CLASS	MATERIAL	TYPE
M1	Agitator	-	1	-	SS316	-	-	-
M2	Top Manhole	ø600	1	6T	SS304	MANHOLE	SS304	-
M3	Side Manhole	-	1	6T	SS304	MANHOLE	SS304	-
N1	Slurry Inlet	DN50	1	10S.	SS304	FLANGE	SS304	AISI150
N2	CIP Spray Ball	DN150	2	10S.	SS304	FLANGE	SS304	AISI150
N3	Water Inlet	DN80	1	10S.	SS304	FLANGE	SS304	AISI150
N4	Slurry Outlet	DN65	1	10S.	SS304	FLANGE	SS304	AISI150
N5	Over Flow	DN80	1	10S.	SS304	FLANGE	SS304	AISI150
N6	Vent	DN80	1	10S.	SS304	FLANGE	SS304	AISI150
N7	Drain	DN150	1	10S.	SS304	FLANGE	SS304	AISI150
N8	Level Transmitter	DN50	1	10S.	SS304	FLANGE	SS304	AISI150
N9	Spare	DN100	1	10S.	SS304	FLANGE	SS304	AISI150
N10	Spare	DN25	1	10S.	SS304	FLANGE	SS304	AISI150
N11	Spare	DN25	1	10S.	SS304	FLANGE	SS304	AISI150
N12	Spare for pH Probe	DN50	1	10S.	SS304	FLANGE	SS304	AISI150

BILL OF MATERIAL

NO.	Q'TY	DESCRIPTION	MATERIAL
1	1	SHELL	PL.6 MM.THK. ASTM A240 TP304
2	1	TOP HEAD	PL.6 MM.THK. ASTM A240 TP304
3	1	BOTTOM HEAD	PL.6 MM.THK. ASTM A240 TP304
4	4	LIFTING LUG	PL.20 MM.THK. ASTM A240 TP304
5	4	BUFFER PLATE	PL.6 MM.THK. ASTM A240 TP304



DETAIL OF N3, N6
SCALE 1:15